



Amay Dixit

Roll No.: 12340220

B.Tech, Data Science and AI Engineering

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GitHub

LinkedIn

EDUCATION

• Indian Institute of Technology, Bhilai

2023 – 2027

B.Tech in Data Science and Artificial Intelligence Engineering (Pursuing)

CGPA: 9.18/10

EXPERIENCE

• RSTUF, OpenSSF, The Linux Foundation (LFX)

May 2026 – Aug 2026

Project Intern — The Update Framework(TUF), Supply Chain Security

- Implemented custom signing-key support and delegation in RSTUF's secure software distribution infrastructure.
- Containerized and integrated the TUF Metadata Visualizer into Kubernetes-based RSTUF deployments using Helm.
- Expanded test coverage and improved documentation to streamline onboarding and adoption of RSTUF.

• Modular Open Source Identity Platform (MOSIP)

Mar 2025 – Feb 2026

Project Intern — Verifiable Credentials, Spring Boot

- Architected a VC revocation mechanism compliant with W3C VCs 2.0, using BitString Status List v1.0 for privacy-preserving, scalable revocation.
- Designed low-latency REST services handling cryptographic proofs (JWT, mDoc) with careful attention to serialization, memory usage, and fault tolerance.

PERSONAL PROJECTS

• HalfLife — Temporal-Aware RAG Reranking Engine [GitHub]

April 2026

Middleware layer for improving retrieval quality in RAG pipelines

- Built a FastAPI service that sits between vector databases (Qdrant) and LLMs, dynamically reweighting retrieval scores using semantic relevance, temporal decay, and trust signals.
- Used Redis for metadata caching and implemented query intent classification (fresh/historical/static) to surface recent, relevant information over older, highly-cited documents.

• ACORDE — Local-First Peer-to-Peer Sync Engine [GitHub]

February 2026

Go-based distributed sync system with CRDT conflict resolution

- Designed a Go-based local-first sync engine using SQLite for on-device storage, CRDTs (LWW registers, OR-Sets) for conflict resolution, and libp2p for transport/peer discovery.
- Embedded per-entry ACLs into the data model for decentralized access control; treated synchronization as a replication problem rather than request-response.

• BitTorrent Client in Go [GitHub]

June 2025

Distributed peer-to-peer file sharing system

- Implemented a full BitTorrent client handling peer discovery, TCP connections, piece scheduling, and concurrent downloads using goroutines and channels.
- Designed integrity verification using SHA-1 hashing and implemented DHT-based peer discovery for fault-tolerant, scalable networking.

TECHNICAL SKILLS

Languages: Go, TypeScript, JavaScript, Python, Java, C++, Rust, HTML/CSS

Frontend: Next.js (App Router), React, Tailwind CSS, Styled Components, TypeScript

Backend: FastAPI, Node.js, Express.js, Spring Boot, Django, Go (Gin)

DevOps & Infra: Docker, Kubernetes, Helm, GitHub Actions, AWS, Nginx, Grafana, Prometheus

Databases: PostgreSQL, Redis, MongoDB, SQLite, Neo4j

ML/AI: LangChain, LLM Integration, RAG Systems, PyTorch, Scikit-learn

LEADERSHIP & ACHIEVEMENTS

• LFX Mentee, RSTUF, OpenSSF, LFX Mentorship

2026

• Captain, AWS Cloud Club, IIT Bhilai

2025–26

• Coordinator, OpenLake — Open Source Club, IIT Bhilai

2025–26

• C4GT DMP Contributor: Selected from 4000+ contributors

May 2025

• Codeforces Specialist: 1432 Rating - Specialist

Jul 2025